📅 Training Day 6 Report

Date: 22 June 2025
Module Name: Network Protocols, VMware OS Setup & Reconnaissance Tools
Mode: Practical + Theoretical (Online)

🔧 Main Focus of the Module

On Day 6, the session covered both **theoretical and hands-on topics**. The first half focused on **essential network protocols** used in communication and hacking. The second half involved setting up different operating systems in **VMware** and executing **practical reconnaissance tools and commands** commonly used in ethical hacking and information gathering.

📚 Topics Covered

🌐 Common Network Protocols in Cybersecurity

Protocol	Port	Description	Usage
HTTP	80	Unsecured web traffic	Web browsing
HTTPS	443	Secure web traffic	Encrypted web services
FTP	21	File Transfer Protocol	Transferring files
SSH	22	Secure Shell	Remote login & command execution
DNS	53	Domain Name Resolution	Converts domain to IP
SMTP	25	Email sending	Email communication
SNMP	161	Simple Network Management	Device monitoring
Telnet	23	Remote login (insecure)	Deprecated; replaced by SSH
ICMP	N/A	Ping, traceroute	Connectivity check

💻 VMware Setup and OS Installations

Operating Systems Installed:

- Kali Linux
- Windows 10

- Windows 11
- Ubuntu

Steps Followed:

1. Created virtual machines in VMware Workstation
2. Configured memory, disk space, and network type (NAT/Bridged)
3. Installed OS using ISO files
4. Performed post-installation setup:
   - Updated OS
   - Installed VMware Tools
   - Verified network using `ping` and `ifconfig`

🔧 Basic Linux Networking & Privilege Commands

Command	Purpose
`ifconfig` / `ip a`	Show IP and interface details
`ping [IP/domain]`	Check connectivity
`sudo`	Run command as superuser
`su -`	Switch to root user
`apt update && apt upgrade`	Update packages
`clear`	Clear terminal screen

🕵 Reconnaissance Tools Used

1. `recon-ng`

- A full-featured reconnaissance framework similar to Metasploit
- Installed using:

```
bash
Copy code
git clone https://github.com/lanmaster53/recon-ng.git
cd recon-ng
pip install -r REQUIREMENTS
```

2. `netcraft`

- Website: https://www.netcraft.com
- Used to gather site profile, hosting, technologies, and attack surface.

3. `dmitry` (Deepmagic Information Gathering Tool)

- Basic information gathering tool
- Command used:

```
bash
Copy code
dmitry -winsep -o output.txt [target.com]
```

4. `sherlock`

- Finds usernames across many websites
- Installed using:

```
bash
Copy code
git clone https://github.com/sherlock-project/sherlock.git
cd sherlock
python3 sherlock.py [username]
```

5. `Maltego`

- Graph-based reconnaissance tool
- Installed via .deb file or GUI
- Used for mapping relationships between people, domains, IPs

6. `Sublist3r`

- Subdomain enumeration tool
- Command:

```
bash
Copy code
python3 sublist3r.py -d example.com
```

7. `ShellGPT`

- AI CLI assistant using OpenAI API
- Installed using:

```
bash
Copy code
pip install shell-gpt
sgpt --help
```

? Hands-On Practice Summary

- Verified interface connectivity with `ifconfig` and `ping`
- Switched to root user using `sudo su -`
- Installed multiple recon tools via `git clone`
- Used recon tools on sample domains like `example.com` and `target.org`
- Compared manual vs automated intelligence gathering

Key Takeaways

- Understood how each protocol is targeted or protected during attacks
- Gained confidence in using terminal commands in Kali Linux
- Learned to install and use multiple open-source reconnaissance tools
- Realized the importance of environment setup (VMware, OS, networking) in ethical hacking labs